Tô Gia Bảo

Data Engineer

Thu Duc, Ho Chi Minh City, Vietnam

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Summary

Data Engineer in digital banking with experience implementing and operating ETL pipelines using Python, SQL, and AWS. Hands-on with data lakes, dimensional modeling, PII masking, automated testing, and operational reporting. Optimized storage for an S3 table by 45% and delivered dbt models and dashboards for monitoring data platform health.

Experience

Timo Digital Bank by BVBank — *Ho Chi Minh City, Vietnam*

Data Engineer — *Mar. 2025 – Present*

- Implemented MariaDB ingestion workflows with Amazon EMR and AWS Glue, applying PII masking in the curated layer and dbt transformations to deliver analytics-ready datasets.
- Optimized storage for one S3 table through Parquet ZSTD-3 compression and file compaction, saving 38.4 TB (45%) and reducing object count by 80% across 56.8 billion rows (2023 to Sep. 2026).
- Developed dbt fact and dimension models with defined grains, keys, and relationships, powering Lightdash dashboards for Airflow reliability, SLA tracking, Athena usage, and S3 storage health.
- Used dbt tests, Great Expectations, and pytest to validate data transformations and integrations, including regression checks for query contracts, SQL compilation, and dbt/Lightdash metadata consistency.
- Developed a FastMCP server powering conversational analytics in Slack, grounded in dbt docs, Lightdash dashboards, and semantic-layer metadata for self-service data discovery.
- Automated MongoDB backup validation and general ledger lookups with Python and Airflow, reducing manual checks.
- Migrated Timo Metrics from Loki/Grafana to SigNoz, centralizing logs and metrics to support operational monitoring and troubleshooting.
- Delivered Tracking247 to sync Customer-provided shipment IDs and status updates into MongoDB via Airflow and Streamlit.

Projects

Mini Lakehouse — *GitHub*

Lakehouse & Platform Engineering — *Jul. 2026 – Present*

- Built a contract-driven AWS lakehouse on S3, Iceberg, Glue, Athena, and EMR Serverless, with Airflow and platform services self-hosted on OCI and infrastructure managed through Terraform.
- Delivered batch and streaming pipelines for GitHub activity, arXiv research, and SSI market data, with dbt transformations, Lightdash and Sigma BI dashboards, Modal GPU OCR, and Netdata monitoring via Cloudflare Zero Trust.

VN Market Pulse — *GitHub*

AI Research & Content Platform — *Jul. 2026*

- Built a Vietnamese research pipeline with multi-source search, deduplication, and reranking to generate source-backed posts.
- Used PydanticAI for typed output and validation, with retrieval budgets, retries, usage tracking, and CLI/Streamlit interfaces.

Technical Skills

Programming & Processing	Python, SQL, Spark/PySpark, Databricks, Amazon EMR
Storage & Modeling	Amazon S3, Apache Iceberg, PostgreSQL, MariaDB, MongoDB; dimensional modeling
Pipelines & Analytics	Airflow, dbt, AWS Glue, Azure Data Factory (ADF), Amazon Athena, Lightdash
Data Quality & Testing	dbt tests, Great Expectations, pytest
DevOps & Observability	Terraform, Docker, Git, SigNoz, Netdata

Education

University of Science, VNU-HCM (HCMUS) — *Ho Chi Minh City, Vietnam*

Bachelor of Data Science | GPA: 3.78/4.00 — *2022 – 2026*

Awards & Certifications

2025 **Top 25**, AIC Challenge
2025 **Recipient**, Ninety Eight Scholarship
2024 **Top 10**, RMIT Data Analytics Competition
2024 **Google Business Intelligence**, Professional Certificate
2024 **Google Data Analytics**, Professional Certificate

Publication

GalaxyAssistant: An Intelligent Assistant for Multimedia Event Retrieval — *SOICT*

Co-author — *2025*

Languages

English TOEIC Listening & Reading: 910 | TOEIC Speaking & Writing: 350